



Lecture 1: Framing Safety and Assurance

CS120: Introduction to AI Safety – September 22, 2026, Zachary Robertson

Instructor: Hi, I'm Zach



Ph.D. candidate, Stanford Computer Science

Research: evaluating AI without ground truth – measurement, incentives, and auditability

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Course assistant: Milan



Ph.D. student, Stanford Computer Science

Research: reliable, adaptive AI systems for open-world physical autonomy

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Logistics

Course webpage: stanford-cs120.github.io/fall2026

Ed: Course questions, announcements, and reading posts

Gradescope: Homework, project check-ins, and discussion tickets

Office hours: Wednesdays, 2:00-3:00 p.m.; Gates 315

Sensitive matters: zroberts@stanford.edu

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How the course works

Before class: Readings and short written responses

During class: Lecture, then discussion or a group activity

Discussion ticket: Keep your sheet; submit its code and your responses

Project: Mid-quarter plan, final project, and two peer reviews

Grading breakdown and full policies: [course webpage](#)

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What comes to mind when you hear "safe AI"?

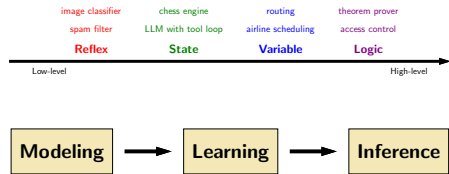
Take one minute to write three associations.

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A course about AI safety

We adapt CS 221's basic paradigm: representations and their instantiation as artifacts.

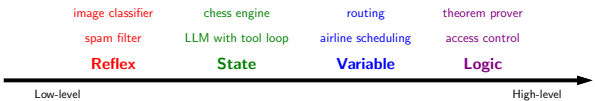


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Four levels of expressiveness

The appropriate assurance depends on the expressiveness of the representation.

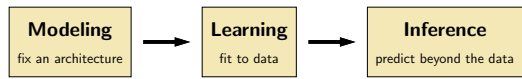


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Machine learning and generalization

If done properly, the model makes usable inferences beyond the training examples.



This is generalization.

Statistics and probability theory formalize it.

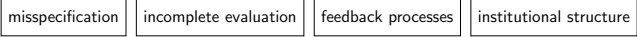
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Assurance beyond generalization

Successful generalization is not taken as the end of the story.

Optimization pressure in increasingly capable systems exploits:



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Evaluation is itself an object of optimization

Classical safety asks whether a system is acceptable.

This course also asks when the process for deciding acceptability becomes an object of optimization.

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Failure, hazard, risk, and harm

Failure: a component or process that departs from its intended behavior

Hazard: a system condition or mechanism that is a source of potential harm

Risk: the likelihood and severity of harm arising from a hazard under operating conditions

Harm: a realized outcome that is negative to interests

These terms are distinguished in causal order, but the sequence is not mandatory.

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Running example: a content filter

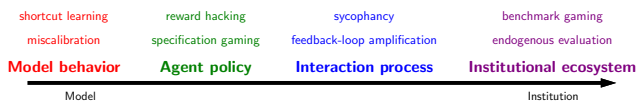
A content filter trained on crowd-sourced toxicity labels.

- Failure:** shortcut learning that fits dialect as a proxy for toxicity
- Hazard:** the resulting spurious correlation in the deployed filter
- Risk:** depends on dialect prevalence, deployment scale, and the impact of wrongful filtering
- Harm:** systematic filtering of content from speakers of that dialect

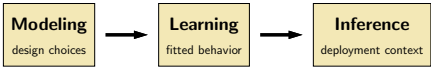
The recurring question

What must be assured for this unit of analysis, and what can optimization pressure exploit?

Optimization pressure by unit of analysis



Where hazards can enter



Hazards can enter at any stage. Whether harm occurs depends on how the system is used and under what conditions.

Four modules and their units of analysis

	Typical systems
2 Model behavior	classifier, content filter
3 Agent policy	chess engine, robot planner, tool-using LLM
4 Interaction process	recommender, team coordinator, assistant
5 Institutional ecosystem	platform market, regulator, benchmark organization

Hazards by module

	Hazards
2 Model behavior	spurious correlations, untested inputs, evaluation blind spots
3 Agent policy	proxy rewards, world-model errors, long horizons
4 Interaction process	human bias, strategic responses, non-stationarity
5 Institutional ecosystem	metric definitions, evaluator dependence, jurisdiction gaps

Each entry is a system condition or mechanism that can be a source of potential harm.

Potential harms by module <table> <tr> <td></td> <td>Potential harms</td> </tr> <tr> <td>2 Model behavior</td> <td>wrongful filtering, unequal error burdens, unsafe decisions</td> </tr> <tr> <td>3 Agent policy</td> <td>irreversible side-effects, resource loss, policy violations</td> </tr> <tr> <td>4 Interaction process</td> <td>loss of autonomy, entrenched inequities, distorted decisions</td> </tr> <tr> <td>5 Institutional ecosystem</td> <td>weakened oversight, concentrated power, systemic externalities</td> </tr> </table> Each entry names an outcome that would be negative to interests if realized. CS120 18		Potential harms	2 Model behavior	wrongful filtering, unequal error burdens, unsafe decisions	3 Agent policy	irreversible side-effects, resource loss, policy violations	4 Interaction process	loss of autonomy, entrenched inequities, distorted decisions	5 Institutional ecosystem	weakened oversight, concentrated power, systemic externalities	The content filter across four units of analysis Model behavior: the classifier fits dialect as a shortcut for toxicity Agent policy: the pipeline removes, warns, or escalates based on classifier outputs Interaction process: users rephrase to evade the filter, and the input distribution shifts Institutional ecosystem: the platform reports accuracy using a metric it defines CS120 19
	Potential harms										
2 Model behavior	wrongful filtering, unequal error burdens, unsafe decisions										
3 Agent policy	irreversible side-effects, resource loss, policy violations										
4 Interaction process	loss of autonomy, entrenched inequities, distorted decisions										
5 Institutional ecosystem	weakened oversight, concentrated power, systemic externalities										
Units of analysis as perspectives Each unit of analysis is a perspective. The modules are neither exclusive categories nor hierarchical. An RL agent may use reflexes to navigate. An assurance question will not always reduce to one module. CS120 20	Two further questions for Module 1 Meeting 2: What is the object of assurance? Meeting 3: What counts as an assurance claim? CS120 21										
Course plan Module 1: framing safety and assurance topics: failure, hazard, risk, harm; object of assurance; what counts as an assurance claim arc: vocabulary and the recurring question; three meetings Module 2: model behavior topics: shortcut learning, adversarial vulnerability, miscalibration, distribution shift arc: development-time optimization; classifier and content-filter case studies Module 3: agent policy topics: reward hacking, specification gaming, unsafe exploration, side-effects arc: sequential decisions and agency; planners and tool-using LLMs Module 4: interaction process topics: sycophancy, manipulation, preference shaping, feedback-loop amplification arc: humans enter the loop; recommenders and assistants Module 5: institutional ecosystem topics: benchmark gaming, endogenous evaluation, regulatory capture arc: evaluation itself becomes an object of optimization CS120 22	Discussion: a content filter A platform recently implemented a content filter using a classifier trained on crowd-sourced toxicity labels. The platform has observed a large increase in account-suspension appeals. The toxicity classifier appears to be triggering automatic account suspensions. Discuss with classmates; record your own responses on your ticket. CS120 23										

Discussion: apply the vocabulary

Identify one possible failure, hazard, or harm. Label the category and explain briefly. What operating conditions would you need to clarify to assess the risk?

Short answers are sufficient. Make assumptions explicit.

Discussion: test an assurance inference

Suppose the classifier performs well on held-out toxicity labels. What does that establish? What does it leave unresolved about the increase in appeals?

Separate the measurement from the assurance claim.

Discussion: record a revision or clarification

After discussion, identify one assumption to clarify or one revision to make. Explain why.

Write your own final response before submitting.

Submit your discussion ticket

Q1: Enter the code printed on your September 22 ticket.

Q2: Type your responses or upload a readable photo or scan.

Deadline: Use the due time shown in the Gradescope assignment.

One student per code. Keep the same code if you resubmit.

Questions and next meeting

September 24: What is the object of assurance?

Concrete Problems in AI Safety
Amodei et al., 2016

Fairness and Abstraction in Sociotechnical Systems
Selbst et al., 2019

Course website and Ed: readings and response instructions